Lab Assignment (C#): Monte Carlo methods

Application to the Ising model

Computational Physics Course

1. Theoretical introduction

Ferromagnetic materials, such as iron or nickel, are composed of atoms whose free electrons possess a permanent magnetic moment, even in the absence of an external magnetic field. The most paradigmatic model used to study this system is the Ising model; moreover, it is one of the few models for which an exact solution is known in one and two dimensions, playing a fundamental role in the understanding of ferromagnetic materials and phase transitions. The Ising model is defined on a regular lattice that mimics a crystalline structure, where at each site a magnetic moment or spin s_i is placed, which can take the values $+1, -1$, *spin up or spin down*. The Ising model is a model of interacting particles in which the interaction occurs between nearest neighbors, that is, short-range interactions are considered. In this way, given the spin s_i, this site contributes to the total energy of the system through

$$H_i = -J \sum_{\langle ij \rangle} s_i s_j, \tag{1.1}$$

where $\langle ij \rangle$ denotes the nearest neighbors of site i, and $J > 0$ is the interaction constant. In a one-dimensional (1D) lattice, the contribution H_i takes the following form

$$H_i = -J s_i (s_{i-1} + s_{i+1}), \tag{1.2}$$

while in two dimensions (2D), the contribution of $s_{i,j}$ is written as

$$H_{i,j} = -J s_{i,j} (s_{i-1,j} + s_{i+1,j} + s_{i,j+1} + s_{i,j-1}). \tag{1.3}$$

In both cases, we observe that in 1D the spin interacts with its *left* and *right* neighbors, while in 2D we must add to the previous sum the *up* and *down* neighbors.
With the above in mind, we can write the total energy of the system by summing the contribution of all spins, that is:

$$H_{\{s_i\}} = \frac{1}{2} \sum_i H_i, \tag{1.4}$$

where the index i runs over all the spins in the system, and the factor $1/2$ is introduced as a consequence of counting each contribution twice.

As mentioned previously, this system contributed significantly to the understanding of phase transitions. To clarify this, we assume that the system is in thermal equilibrium at temperature T. If we were to consider only the energy, the system would evolve by minimizing it, and the resulting phase would be perfectly ordered. Indeed, from expression (1.1), the terms $s_i s_j$ would tend to take the same value, meaning that all spins would tend to align with their nearest neighbors. However, there is also the effect of temperature: thermal fluctuations allow spins to randomly change their orientation. This effect becomes more significant as the temperature increases; therefore, high temperatures give rise to disordered phases. Depending on the strength of the temperature effect, the system may exhibit ordered or disordered phases, thus reflecting a competition between energy minimization and thermal fluctuations. To quantify the degree of order in the system, we introduce the magnetization per spin m. For a system of N_e spins, the magnetization m is given by the following expression:

$$m = \frac{1}{N_e} \sum_i s_i \tag{1.5}$$

where the sum runs over the N_e spins of the system. We note how this order parameter behaves: if all spins have the same orientation, either all up or all down, we obtain $|m| = 1$. On the other hand, in a completely disordered configuration, the value of $|m|$ fluctuates around zero.

During this exercise, we will study through computational experiments the effect of temperature on a system modeled by the Ising model using the Metropolis algorithm, which will be introduced next.

1.1. Metropolis algorithm

iven our system of N_e spins, we assume that we have the following configuration $s_{i_{i=1}}^{N_e}$. The Metropolis algorithm states that a change in the spin s_i (a flip) occurs with probability

$$P(s_i \to -s_i) = \exp\left(-\frac{\Delta H}{k_B T}\right), \tag{1.6}$$

where k_B is the Boltzmann constant and ΔH is the energy change between the original system and the system after flipping the spin. The computation of this energy difference is straightforward. Suppose we flip the spin s_i, so that it takes the new value $s_i' = -s_i$. The

resulting change in energy, using expressions (1.1) and (1.4), is given by:

$$\begin{aligned}\Delta H = H_i' - H_i &= \left(-Js_i' \sum_{\langle ij \rangle} s_j\right) - \left(-Js_i \sum_{\langle ij \rangle} s_j\right) \\ &= \left(+Js_i \sum_{\langle ij \rangle} s_j\right) + \left(Js_i \sum_{\langle ij \rangle} s_j\right) \\ &= 2Js_i \sum_{\langle ij \rangle} s_j.\end{aligned} \tag{1.7}$$

Taking the above into account, we write the Metropolis algorithm as follows. We assume that initially the system is in a completely random phase, where the spins have been assigned up or down at random. Starting from this initial configuration:

1. A spin s_i is chosen at random from the system. This spin is the candidate to be flipped.
2. The quantity ΔH is computed using expression 1.7. This gives the energy change associated with flipping the spin s_i.
3. If $\Delta H < 0$, the change $s_i \to -s_i$ is accepted immediately.
4. Otherwise, a random number r uniformly distributed in the interval $[0,1)$ is generated.
5. If $r < \exp(-\Delta H/(k_B T))$, the change $s_i \to -s_i$ is accepted.
6. Otherwise, the system remains unchanged.

Regarding this algorithm, we note that since the function $\exp(-\Delta H/(k_B T))$ increases with temperature, the condition given in step 5 will be satisfied more often as the value of T increases, thus introducing thermal fluctuations. On the other hand, this algorithm must be repeated for a sufficient number of steps in order for the system to reach thermal equilibrium; in this state, the magnetization becomes constant apart from small fluctuations.

Finally, in this type of simulations it is common to impose periodic boundary conditions on the system in order to avoid undesired boundary effects. We consider a one-dimensional spin lattice of size L, i.e., our system is given by L ordered spins $s_i i = 1^L$. For a spin s_i with $1 < i < L$, its energy is given in terms of the contributions $s_i si - 1$ and $s_i s_{i+1}$. Imposing periodic boundary conditions means that for $i = 1$, the *left* neighbor is s_L, while the right neighbor is s_2. Applying analogous reasoning for $i = L$, we obtain that under periodic boundary conditions:

$$\begin{cases} \Delta H_1 = 2Js_1(s_L + s_2) & \text{si } i = 1. \\ \Delta H_i = 2Js_i(s_{i-1} + s_{i+1}) & \text{si } 1 < i < L. \\ \Delta H_L = 2Js_L(s_{L-1} + s_1) & \text{si } i = L. \end{cases} \tag{1.8}$$

This type of boundary condition can be extended to 2D by applying it independently in each dimension. As an example, for the spin $s_{1,1}$, its contribution under periodic boundary conditions on an $L \times L$ lattice is given by:

$$\Delta H_{1,1} = 2Js_{1,1}\left(s_{1,L} + s_{1,2} + s_{L,1} + s_{2,1}\right). \tag{1.9}$$

2. Questions to be addressed

Throughout this exercise, we will take the interaction constant as $J = 1$ and express the temperature in units of k_B, i.e., we will also set $k_B = 1$. In this way, when we write $T = s$, where s is any positive value, we are effectively stating that $Tk_B = s$. On the other hand, the variable associated with the number of steps towards equilibrium, N_P, will be assumed to have the form $N_P = 10^A$, where the exponent A is an input parameter of the simulation. Furthermore, in each realization of the model we will initialize the system randomly as follows. For each site of the system, the variable s_i takes the value $+1$ with probability $1/2$, and therefore the probability of the state -1 is equal to that of the state $+1$. This is implemented by generating a uniformly distributed random number in the interval $[0,1)$ for each site of the lattice. If at site i the value is less than 0.5, then s_i is set to $+1$; otherwise, it is set to -1. With these initial considerations, proceed to complete the following sections:

1. Implement a program simulating the two-dimensional Ising model with periodic boundary conditions on a square lattice of size $L \times L$. The program must take as input arguments the system size L, the exponent A associated with the number of steps towards equilibrium, and the temperature T. For each simulation, the output should be a data file with L rows and L columns such that the entry corresponding to row i and column j is the value $s_{i,j}$. Obtain this file for the following parameters and represent the resulting data as a heat map using two colors, red and blue, corresponding to each of the two possible spin orientations:

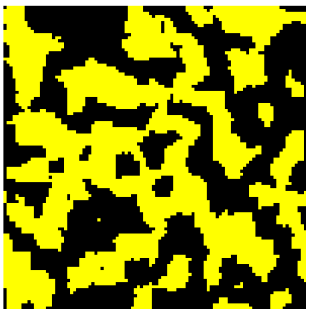
a) $L = 100, A = 5$ y $T = 1$.

b) $L = 100, A = 8$ y $T = 1$.

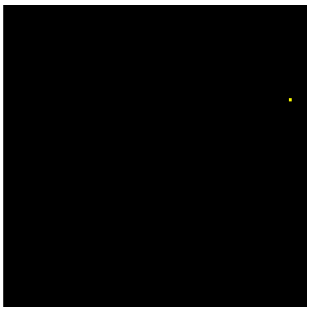
c) $L = 100, A = 8$ y $T = 2$.

d) $L = 100, A = 8$ y $T = 2.5$.

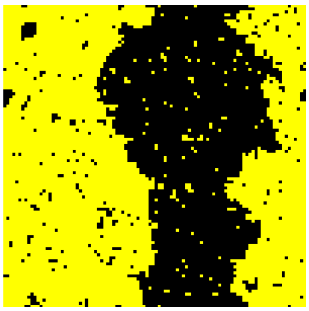
Solution



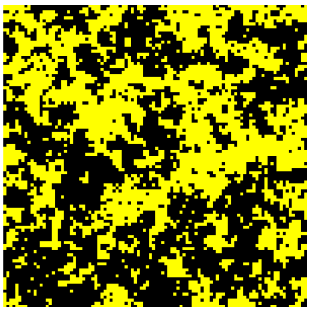
(a) $L = 100; A = 5; T = 1$



(b) $L = 100; A = 8; T = 1$



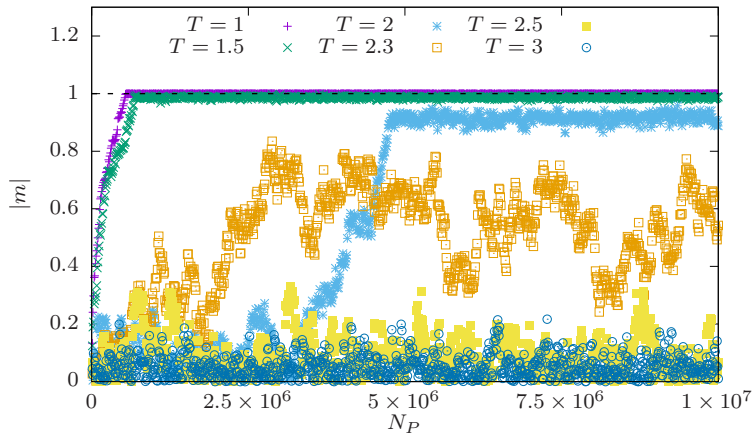
(c) $L = 100; A = 8; T = 2$



(d) $L = 100; A = 8; T = 2.5$

2. As explained previously, the Ising model undergoes a transition from an ordered phase to a disordered phase. This phase transition occurs at a specific temperature, known as the critical temperature, which takes the value $T_c = 2.27$. As can be observed from the results of the previous section, below the critical temperature the system remains ordered, whereas for temperatures above the critical temperature the system becomes disordered. The degree of order is measured through the magnetization per spin m, whose expression is given in Eq. (1.5). This order parameter evolves as a function of the number of steps towards equilibrium, and therefore a sufficiently large number of steps N_P is required.
In this section, we aim to determine whether, for a given value of N_P, the magnetization has reached equilibrium. To this end, we let the system evolve by initially setting the number of steps to $N_P = 10^7$, and we record the value of the magnetization every 10^4 steps. Carry out this task by modifying the previous program, so that instead of outputting the spin configuration matrix, it produces an output file consisting of a matrix with 2 columns and 10^3 rows, where the first column corresponds to the number of steps and the second column corresponds to the magnetization of the system at that time. Perform these measurements for a square lattice of size 50×50 ($L = 50$) and for the following set of temperatures: $T = 1, 1.5, 2, 2.3, 2.5, 3$. Finally, plot the magnetization as a function of the number of steps for all the requested temperatures in a single graph.

Solution



3. Onsager obtained the exact solution for the two-dimensional Ising model in 1944. In particular, the magnetization as a function of temperature is given by the following expression:

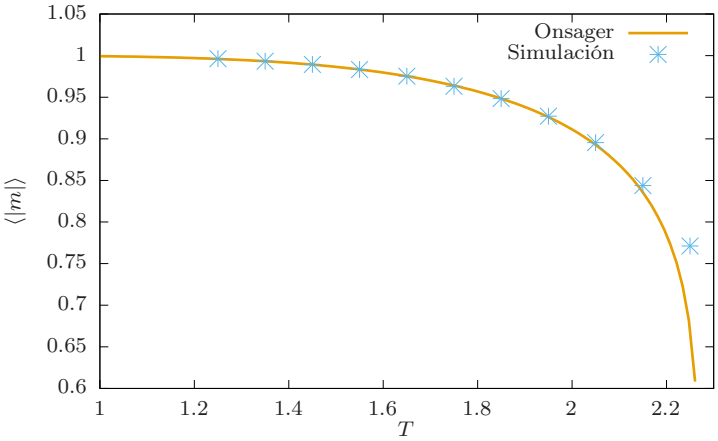
$$|m| = \left(1 - \sinh\left(\frac{2J}{T}\right)^{-4}\right)^{1/8}. \tag{2.1}$$

On the other hand, as observed in the previous section, the magnetization still fluctuates even at equilibrium. For this reason, in this type of method, namely Monte Carlo methods, fluctuations are reduced by averaging over multiple realizations of the system. In this sense, it is common to perform N realizations of the system, obtaining N different values of the magnetization at equilibrium, and taking the value of m as the average over these N realizations, that is:

$$\langle |m| \rangle = \frac{1}{N} \sum_{i=1}^{N} |m_i|,$$

where m_i is the magnetization obtained in the i-th realization of the system after N_P steps. In this section, we will experimentally verify Onsager's solution 2.1. To this end, simulate the two-dimensional Ising model on a square lattice of size 10×10 ($L = 10$). In this simulation, we aim to compute $\langle |m| \rangle$ using $N = 5000$ realizations of the system, where the value of $|m_i|$ is obtained after $N_P = 10^5$ steps. This procedure is carried out over the temperature range $[1.25, 2.25]$ with a step size $\delta T = 0.1$, thereby obtaining 11 points of the curve $\langle |m| \rangle (T)$. Finally, plot these points together with the exact Onsager solution.

Solution


Onsager
Simulación
1.05
1
0.95
0.9
0.85
0.8
0.75
0.7
0.65
0.6
$\langle |m| \rangle$
1
1.2
1.4
1.6
1.8
2
2.2
T